Haithem Turki

Curriculum Vitae

Updated October, 2026
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Education

Carnegie Mellon University **2019 - 2024**
Ph.D. in Computer Science. Thesis: Towards City-Scale Neural Rendering
Advisor: Deva Ramanan. Committee: Jessica Hodgins, Martial Hebert, Jonathan Barron

Stanford University **2012 - 2018**
M.S. in Computer Science

Stanford University **2008 - 2012**
B.S. in Computer Science, minor in Management Science and Engineering

Experience

NVIDIA Research, Senior Research Scientist **2024-**
Spatial Intelligence Lab (w/ Zan Gojcic, Sanja Fidler) → AI-Mediated Reality and Interaction group (w/ Shalini De Mello)

Meta, Research Scientist Intern w/ Christian Richardt, Michael Zollhöfer **2023**
Argo AI, Research Intern w/ Deva Ramanan, Francesco Ferroni **2022**
Palantir Technologies, Engineering Manager and Technical Lead **2012 - 2019**
Addepar, Engineering Intern **2011**
Stanford University, Research Assistant w/ Daphne Koller, M. Pawan Kumar **2010**

Awards & Honors

NeurIPS Oral (top 0.4% of submissions) for "Déjà View" **2026**
ECCV Outstanding Reviewer Award **2026**
CVPR Best Paper Award Candidate for "DIFIX3D+" **2025**
CVPR Highlight (top 2.8% of submissions) for "HybridNeRF" **2024**
CVPR Highlight (top 2.8% of submissions) for "SpecNeRF" **2024**

Software & Product Impact

NVIDIA Omniverse NuRec, core contributor — neural reconstruction for robotics and AV simulation (Isaac Sim, CARLA, AlpaSim) 2024 - 2026
Shipped components include Fixer/Harmonizer [13, 16], LiDAR simulation [15], and Asset Harvester (Technical Report [2])

ArtiFixer, code and model checkpoints for [17] (691 GitHub stars) 2026
Déjà View (DVLT), code and model weights for [18] (476 GitHub stars) 2026
SimULi, LiDAR/camera simulation for [15], in nerfstudio gsplat 2026
SUDS, code for [7] (199 GitHub stars) 2023
PyNeRF, Nerfstudio plugin for [10] (61 GitHub stars) 2023
Mega-NeRF, code, models, and Mill 19 dataset for [5] (504 GitHub stars) 2022

PUBLICATIONS

* equal contribution † equal senior authorship

[18] A. Burzio*, T. Fischer*, S. Elflein, Q. Zhou, R. de Lutio, J. Ren, J. Huang, S. Huang, M. Pollefeys, L. Leal-Taixé, Z. Gojcic†, **H. Turki**†. "Déjà View: Looping Transformers for Multi-View 3D Reconstruction." NeurIPS 2026 ***(Oral)***

[17] R. de Lutio*, T. Fischer, Y.-Y. Chang, Y. Zhang, J. Z. Wu, X. Ren, T. Shen, K. Tothova, Z. Gojcic, **H. Turki***. "ArtiFixer: Enhancing and Extending 3D Reconstruction with Auto-Regressive Diffusion Models." SIGGRAPH 2026 ***(Invited Oral, GenRecon3D Workshop @ CVPR 2026)***

[16] Y. Zhang*, K. Tothova*, Z. Wang, K. Yin, **H. Turki**, R. de Lutio, Y.-Y. Chang, O. Litany, S. Fidler, Z. Gojcic. "DiffusionHarmonizer: Bridging Neural Reconstruction and Photorealistic Simulation with Online Diffusion Enhancer." CVPR 2026

[15] **H. Turki***, Q. Wu*, X. Kang, J. Martinez Esturo, S. Huang, R. Li, Z. Gojcic, R. de Lutio. "SimULi: Real-Time LiDAR and Camera Simulation with Unscented Transforms." ICLR 2026

[14] S. Bahmani, T. Shen, J. Ren, J. Huang, Y. Jiang, **H. Turki**, A. Tagliasacchi, D. B. Lindell, Z. Gojcic, S. Fidler, H. Ling, J. Gao*, X. Ren*. "Lyra: Generative 3D Scene Reconstruction via Video Diffusion Model Self-Distillation." ICLR 2026

[13] Z. Wu*, Y. Zhang*, **H. Turki**, X. Ren, J. Gao, M. Z. Shou, S. Fidler, Z. Gojcic†, H. Ling†. "DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models." CVPR 2025 ***(Best Paper Award Candidate)***

[12] **H. Turki**, V. Agrawal, S. Rota Bulò, L. Porzi, P. Kontschieder, D. Ramanan, M. Zollhöfer, C. Richardt. "HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces." CVPR 2024 ***(Highlight)***

[11] L. Ma, V. Agrawal, **H. Turki**, C. Kim, C. Gao, P. Sander, M. Zollhöfer, C. Richardt. "SpecNeRF: Gaussian Directional Encoding for Specular Reflections." CVPR 2024 ***(Highlight)***

[10] **H. Turki**, M. Zollhöfer, C. Richardt, D. Ramanan. "PyNeRF: Pyramidal Neural Radiance Fields." NeurIPS 2023

[9] S. George, **H. Turki**, Z. Feng, T. Eiszler, D. Ramanan, P. Pillai, M. Satyanarayanan. "Low-Bandwidth Self-Improving Transmission of Rare Training Data." ACM MobiCom 2023

[8] S. George, **H. Turki**, Z. Feng, T. Eiszler, D. Ramanan, P. Pillai, M. Satyanarayanan. "Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data." NetAISys Workshop 2023

[7] **H. Turki**, J. Y. Zhang, F. Ferroni, D. Ramanan. "SUDS: Scalable Urban Dynamic Scenes." CVPR 2023

[6] M. Satyanarayanan, Z. Feng, S. George, J. Harkes, R. Iyengar, **H. Turki**, P. Pillai. "Accelerating Silent Witness Storage." IEEE Micro 42(6), 2022

[5] **H. Turki**, D. Ramanan, M. Satyanarayanan. "Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs." CVPR 2022

[4] S. George, T. Eiszler, R. Iyengar, **H. Turki**, Z. Feng, J. Wang, P. Pillai, M. Satyanarayanan. "OpenRTiST: End-to-End Benchmarking for Edge Computing." IEEE Pervasive Computing 19(4), 2020

[3] M. Satyanarayanan, T. Eiszler, J. Harkes, **H. Turki**, Z. Feng. "Edge Computing for Legacy Applications." IEEE Pervasive Computing 19(4), 2020

[2] M. P. Kumar, **H. Turki**, D. Preston, D. Koller. "Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation." IEEE TPAMI 37(7), 2015

[1] M. P. Kumar, **H. Turki**, D. Preston, D. Koller. "Learning Specific-Class Segmentation from Diverse Data." ICCV 2011

Technical Reports & Preprints

[4] X. Huang, T. Li, A. Mazumdar, S. Wang, **H. Turki**, M. Stengel, S. Huang, S. See, K. C. Cheung, R. Wan, S. De Mello. "Trajectory-Anchored Compact Gaussians for Feed-Forward 4D Reconstruction." Under review, 2026

[3] A. Basant, A. Kar, D. Paschalidou, F. Wei, F. Ferroni, G. Garcia Cobo, **H. Turki**, H. Ling, J. Seo, J. Lucas, J. Z. Wu, J. Wang, J. Lorraine, J. Gao, K. He, K. Tothova, K. Xie, M. Tyszkiewicz, Q. Wu, R. de Lutio, R. Li, S. Fidler, S. W. Kim, T. Shen, T. Cao, T. Pfaff, W. Lew, X. Wu, X. Ren, Y. Lu, Y. Zhang, Z. Gojcic, Z. Wang. "NVIDIA OmniDreams: Real-Time Generative World Model for Closed-Loop Autonomous Vehicle Simulation." arXiv:2606.03159, 2026

[2] T. Cao*, J. Ren*, Y. Zhang, J. Seo, J. Huang, S. Solanki, H. Zhang, M. Guo, **H. Turki**, M. Li, Y. Zhu, S. Zhang, Z. Gojcic, S. Fidler, K. Yin. "Asset Harvester: Extracting 3D Assets from Autonomous Driving Logs for Simulation." arXiv:2604.18468, 2026

[1] Z. Feng, S. George, **H. Turki**, R. Iyengar, P. Pillai, J. Harkes, M. Satyanarayanan. "Improving Edge Elasticity via Decode Offload." Technical Report CMU-CS-21-139, Carnegie Mellon University, 2021

Patents

[2] R. Fink, A. Al Mallah, **H. Turki**. "Indexing and Relaying Data to Hot Storage." US Patents 12,481,635 B2 (2025), 11,914,566 B2 (2024), 11,669,507 B2 (2023), 11,176,113 B2 (2021). Pending continuation: US 2026/0093682 A1 (2026).

[1] S. Kromwijk, **H. Turki**, S. Cohen, Y. Xu, F. A. Kara. "Systems and Methods for Constraint Driven Database Searching." US Patents 12,099,509 B2 (2024), 10,915,536 B2 (2021), 10,133,783 B2 (2018).

Teaching

CIS 7000 Neural Scene Representation and Neural Rendering, University of Pennsylvania (Guest Lecture) — 2026

15-640 Distributed Systems, Carnegie Mellon University (Graduate Student Instructor) — 2020 - 2021

17-700 Data Science and Machine Learning at Scale, Carnegie Mellon University (Graduate Student Instructor) — 2020 - 2021

15-821 Mobile and Pervasive Computing, Carnegie Mellon University (Project Mentor) — 2020 - 2021

Student Collaborators & Interns

Tobias Fischer, PhD Student @ ETH Zürich. See Publications [17, 18].

Alessandro Burzio, PhD Student @ University of Modena and Reggio Emilia. See Publications [18].

Yen-Yu Chang, PhD Student @ Cornell University. See Publications [16, 17].

Invited Talks

"Photoreal Worlds at Scale: From Capture to Generation"

USC	Oct 2026
UCLA Computer Science Seminar Series (CS 201)	Oct 2026
EPFL	Jul 2026
ETH Zürich, ETH AI Center Academic Talk Series (AICATS)	Jul 2026
Caltech	Jul 2026
ZurichCV, ETH AI Center	Jun 2026

"Towards City-Scale Neural Rendering"

Waymo Research	Feb 2024
Google	Feb 2024
NVIDIA Toronto AI Lab	Feb 2024
Stanford Vision Lab	Jan 2024

"Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs"

Argo AI	May 2022

Academic Service

Area Chair:
ACCV

Fellowship Reviewer:
NVIDIA Graduate Fellowship Program

Reviewer:
CVPR, ACCV, ECCV, ICCV, NeurIPS, ICLR, SIGGRAPH, SIGGRAPH Asia, AAAI, Transactions on Image Processing

Additional Information

Languages:
English (native), French (fluent), Arabic (fluent), Japanese (elementary), German (elementary)

Entrepreneurship & Industry:
Venture Partner, Page One Ventures (2020 - 2024). Startup mentor and investor, Alchemist Accelerator. Alumnus, Stanford StartX accelerator.